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Segmentation of Any Fire Event (SAFE): A Rapid and High-Precision Approach for Burned Area Extraction Using Sentinel-2 Imagery

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Abstract: The timely and accurate monitoring of wildfires and other sudden natural disasters is crucial for safeguarding the safety of residents and their property. Satellite imagery for wildfire monitoring offers a unique opportunity to obtain near-real-time disaster information through rapid, large-scale remote sensing mapping. However, existing wildfire monitoring methods are constrained by the temporal and spatial limitations of remote sensing imagery, preventing comprehensive fulfillment of the need for high temporal and spatial resolution in wildfire monitoring and early warning. To address this gap, we propose a rapid, high-precision wildfire extraction method without the need for training—SAFE. SAFE combines the generalization capabilities of the Segmentation Anything Model (SAM) and the high temporal effectiveness of hotspot product data such as MODIS and VIIRS. SAFE employs a two-step localization strategy to incrementally identify burned areas and pixels in post-wildfire imagery, thereby reducing computational load and providing high-resolution wildfire impact areas. The high-resolution burned area data generated by SAFE can subsequently be used to train lightweight regional wildfire extraction models, establishing high-precision detection and extraction models applicable to various regions, ultimately reducing undetected areas. We validated this method in four test regions representing two typical wildfire scenarios—grassland and forest. The results showed that SAFE's F1-score was, on average, 9.37% higher than alternative methods. Additionally, the application of SAFE in large-scale disaster scenarios demonstrated its potential capability to detect the fine spatial distribution of wildfire impacts on a global scale.

Keywords: hazard monitor; burned area; segmentation anything model; high-resolution imagery

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1. Background

The profound impact of climate change and human activities on the environment has led to an increased frequency of extreme events such as heatwaves, droughts, dust storms, floods, hurricanes, and wildfires [1,2]. According to the Intergovernmental Panel on Climate Change (IPCC) reports, climate change has significantly intensified the frequency and severity of wildfires, exacerbating their detrimental effects on ecosystems and

human communities [3–5]. Among these events, wildfires have a particularly significant impact on ecosystems, not only destroying infrastructure but also causing casualties [6]. Therefore, the detection of wildfires and the accurate monitoring of their scale and impact over large areas have become increasingly important. In recent years, researchers have introduced a variety of effective wildfire detection systems, including those based on ground and aerial imagery [7,8]. The integration of satellite-based data from systems like MODIS and Sentinel-2 provides comprehensive and high-resolution information that strengthens environmental evaluations by enabling detailed analysis of wildfire behavior and its ecological consequences [9–11]. However, with technological advancement, satellite systems have gradually become an essential means of addressing wildfire issues due to their high reliability, wide coverage, rapid response capabilities, and ability to monitor large areas [12–16]. Satellite systems use thermal sensors and optical sensors to acquire comprehensive data on wildfires, allowing researchers to study wildfire behavior, spread rate, scale, and environmental impact [17]. These data are crucial for informing climate change models and developing strategies to mitigate the environmental impacts of wildfires, thereby providing critical insights into the areas of study and analysis within our research objectives [18,19].

Currently, several global wildfire-burned-area data products are provided by different organizations, playing an important role in large-scale monitoring [20–27]. However, these data products mostly rely on sensors with a high temporal but low spatial resolution (as indicated by the blue sections in Table 1). Although these sensors can frequently capture surface information, enabling the real-time monitoring of wildfire development, their low spatial resolution makes it difficult to detect subtle surface changes, especially for small-scale fires [28]. Additionally, the primary drivers of most fires, such as human activities and climate-induced factors like increased temperatures and prolonged droughts, exacerbate the complexity of managing wildfire events. Existing wildfire management strategies rely heavily on the timely and precise detection of fire-affected areas to implement effective response measures and conduct pre- and post-fire damage assessments [29,30]. Moreover, understanding post-fire recovery and long-term ecological impacts is essential for comprehensive wildfire management [31–33]. Following a wildfire event, ecosystems undergo a series of recovery processes that can vary significantly depending on factors such as fire intensity, frequency, and the specific characteristics of the affected habitat [34,35]. Vegetation regrowth is a primary focus of post-fire studies, with research indicating that fire-adapted species often exhibit rapid regeneration through mechanisms like seed germination triggered by heat or chemical cues from smoke [36]. In contrast, non-adapted species may experience prolonged recovery periods or fail to re-establish, leading to shifts in species composition and ecosystem structure. Integrating burned area data with climate models and land management practices is crucial for predicting and mitigating the long-term impacts of wildfires [37]. The accurate monitoring of burned areas using advanced remote sensing techniques enhances our ability to assess ecological recovery and implement effective restoration strategies.

To address the limitations of high-resolution data in monitoring burned areas, different research teams have collected and compiled burned scar data from multiple historical wildfire events and integrated these datasets into public databases (as indicated by the red sections in Table 1). These datasets significantly improve spatial resolution, typically reaching 30 m, enabling researchers to more accurately identify and analyze the extent and characteristics of burned areas. This progress is particularly important for assessing the impact of wildfires on local ecosystems, analyzing the behavior of small fires, and developing more targeted fire prevention measures [38,39].

Despite the improvement in spatial resolution enhancing the accuracy of wildfire monitoring and enabling a more detailed analysis of burned areas, existing high-resolution datasets are primarily concentrated in a few key regions, such as the United States, Africa, and Brazil [40–42]. This uneven geographic distribution leads to data shortages in certain areas, particularly those with frequent wildfires that have not been adequately studied. Moreover, as spatial resolution improves, the temporal resolution of satellite data often decreases, making the near-real-time monitoring of wildfire dynamics challenging [41]. The lack of real-time data during rapid fire spread may limit the timeliness and effectiveness of emergency responses. Therefore, achieving a balance between high spatial and high temporal resolution in wildfire detection is a critical technical challenge that needs to be addressed.

Table 1. A summary of the name, temporal coverage, spatial resolution, coverage, and sources of datasets/products.

Satellite/Products	Temporal Coverage	Spatial Resolution (m)	Coverage	Source
MODIS Active Fires (MCD14DL)	2000–present	1000	Global	https://firms.modaps.eosdis.nasa.gov (accessed on 16 December 2024)
VIIRS Active Fires	2013–present	375/750	Global	\
VIIRS NightFire (VNF)	2013–present	750	Global	https://eogdata.mines.edu/products/vnf/ (accessed on 16 December 2024)
MCD12Q1	2001–2019	500	Global	https://lpdaac.usgs.gov/products/mcd12q1v006 (accessed on 16 December 2024)
ICNF Burned Areas	1990–2021	Variable	Portugal	https://www.icnf.pt (accessed on 16 December 2024)
MCD64A1	2000–present	500	Global	https://lpdaacsvc.cr.usgs.gov/appeears/ (accessed on 16 December 2024)
FireCCI51	2001–2020	250	Global	https://data.ceda.ac.uk/neodc/esacci/fire/data/ (accessed on 16 December 2024)
LBA_CU	1984–2021	30	The conterminous United States	https://gec.cr.usgs.gov/outgoing/baecv/LBA/ (accessed on 16 December 2024)
FireCCISFD20	2019	20	Sub-Saharan Africa	https://climate.esa.int/en/projects/fire/data/ (accessed on 16 December 2024)

In early burned area extraction algorithms, coarse-resolution satellite images (e.g., MODIS) were frequently used for automatic burned area extraction due to their high revisit rate [43]. These data provide an immediate damage assessment but compromise the spatial accuracy of fire boundaries. Visual interpretation using higher-resolution satellite images (e.g., Sentinel-2 or Landsat) can improve accuracy, but it is time consuming, prone to human error, and difficult to apply on a large scale [44]. With advances in remote sensing technology, fully automated burned scar extraction methods have been developed, using specific spectral indices (e.g., the Normalized Difference Vegetation Index (NDVI) [45], Burned Area Index (BAI) [46], and Normalized Burn Ratio (NBR) [47]) to capture the spectral characteristics of burned areas. Although these index-based methods improve the accuracy of burned area identification to some extent, the complexity of geographic environments can introduce interference, and threshold settings in different studies often rely on human expertise, leading to inconsistent results [48].

With the rapid development of machine learning (ML) technologies, researchers have shown increasing interest in ML-based automated burned area extraction methods, particularly for improving spatial and temporal resolution [49]. ML techniques have the potential to significantly improve the accuracy of burned area extraction in a cost-effective manner. Simple and effective models like Random Forest (RF) have been widely used for burned area extraction [50–54]. However, existing ML methods primarily rely on spectral

features and the temporal information of individual pixels, making them susceptible to noise and limiting their performance in highly heterogeneous regions due to the insufficient utilization of geospatial autocorrelation [55].

In contrast, deep learning methods show great potential in wildfire monitoring due to their powerful spatial information extraction capabilities [56]. Deep-learning-based wildfire area extraction methods typically use U-Net architectures to achieve precise burned area extraction through pixel-wise classification [57] (as shown in Figure 1a). Deep learning is also widely used in change detection, capturing surface changes by comparing multi-temporal images before and after a fire to identify the impact of wildfires [38,57–60] (as shown in Figure 1b). Despite the significant progress made in wildfire extraction, these models are often limited to specific regions due to the lack of sufficient training samples, making them difficult to generalize to larger scales or different geographic regions [61]. Some studies have attempted to address this issue through techniques such as transfer learning and domain adaptation, but data scarcity remains the primary bottleneck limiting model performance [41]. The Segmentation Anything Model (SAM) represents a significant advancement in the field of image segmentation. Developed by Meta AI, SAM is a foundation model that leverages deep learning to perform promptable segmentation across a wide variety of objects and scenes without the need for task-specific training. Its architecture is designed to generalize effectively across different domains, making it highly applicable to diverse remote sensing tasks. In remote sensing studies, SAM has been utilized for applications such as land cover classification, object detection, and change detection, demonstrating a robust performance in handling complex and high-resolution imagery [62–65].

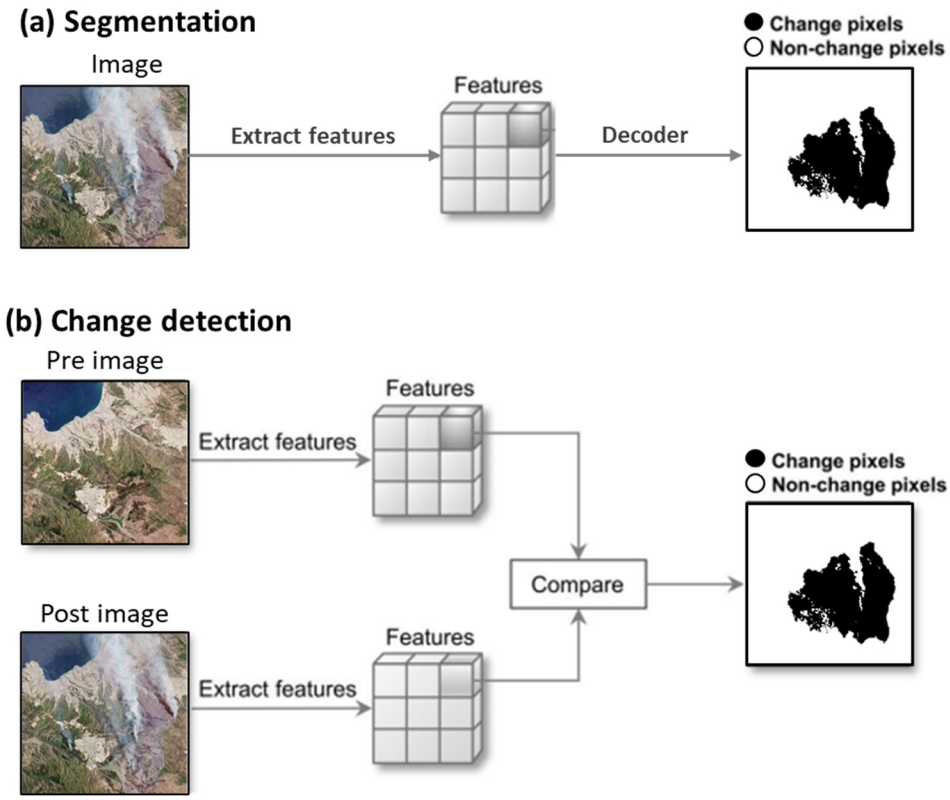


Figure 1. Different types of deep learning methods for burned area detection.

Currently, wildfire area extraction mainly relies on multi-temporal remote sensing images [66–70]. Although this approach identifies burned areas by comparing pre- and post-fire images, surface environment changes, atmospheric conditions, and variations in sensor imaging angles can lead to non-fire-related changes between images, affecting the extraction accuracy [66]. Therefore, efficiently detecting burned areas using single-scene images has become an important research challenge. Compared to multi-temporal image analysis, single-scene image analysis does not rely on temporal sequence data, reducing errors caused by differences in imaging conditions, but it requires more sophisticated algorithms to accurately identify burned areas from limited information [71].

To address these challenges, this paper proposes an integrated framework, Segmentation of Any Fire Event (SAFE), for wildfire area detection and near-real-time monitoring. This framework combines Sentinel-2 optical data with MODIS MCD64A1 data to identify potential fire areas and uses a deep-learning-based Segmentation Anything Model (SAM) to precisely detect and segment fire spots within these areas. For ambiguous or uncertain areas, we designed a semi-supervised extraction method to achieve the rapid identification and extraction of potential burned areas. Furthermore, the high-resolution burned area data generated by SAFE were used to fine-tune deep learning models, improving their applicability across various geographical regions. By leveraging this high-resolution data, the model can enhance its generalization capabilities, allowing it to perform well in regions with limited training data. Notably, our model does not require training or localized parameter adjustment and can be directly applied to burned area extraction in different geographic regions. This integration with deep learning techniques and transfer learning approaches further improves the model's performance in areas with sparse or uneven data distribution, demonstrating its ability to adapt to diverse fire regimes and landscape conditions. This generalization capability ensures that the model maintains high efficiency and accuracy in diverse environments.

2. Methodology

This study proposes the SAFE model, which incorporates the technical framework of SAM 2, aiming to design a modular system for extracting high-resolution burned areas. This section provides a detailed description of the methodology from three aspects: dataset preprocessing, the two-step localization strategy of SAFE, and localized post-processing.

2.1. Dataset Preprocessing and Collection

2.1.1. Burned Area from MCD64A1

MCD64A1 is a global burned area dataset provided by NASA, derived from the Moderate Resolution Imaging Spectroradiometer (MODIS) sensor [72]. The MCD64A1 dataset uses daily surface reflectance and thermal anomaly data from MODIS to detect global burned areas, serving as the primary output of the MODIS burned area product, which provides spatial information on burned areas following fire events.

The dataset mainly draws from data collected by MODIS sensors onboard the Terra and Aqua satellites, which acquire global multispectral data daily, featuring a high temporal resolution. The spatial resolution of MCD64A1 is 500 m, which, despite being relatively low, provides consistent and long-term (2000–present) fire monitoring data, suitable for long-term fire trend analysis (as shown in Table 2).

The extraction algorithm for MCD64A1 adopts a hybrid classification approach, combining spectral change and thermal anomaly information to identify burned areas. Specific processing steps include (1) preprocessing: preprocessing the MODIS daily surface reflectance and thermal anomaly data, including cloud removal and noise filtering, to enhance data quality; (2) spectral change detection: detecting possible burned areas by

comparing surface reflectance at different times to identify vegetation changes; (3) thermal anomaly detection: confirming the occurrence of fires based on thermal anomaly data combined with spectral changes; (4) final classification: classifying detected burned areas into monthly data products and generating raster files.

Table 2. Overview of multi-source datasets used in the SAFE framework.

Dataset Name	Spatial Resolution	Temporal Coverage	Data Sources	Spectral Bands of Sentinel-2
MODIS MCD64A1	500 meters	2019–2023	NASA	N/A
Sentinel-2 MSI	10 meters	2015–2023	ESA	Band 2 (Blue), Band 3 (Green), Band 4 (Red), Band 6 (Red Edge2), Band 7 (Red Edge3), Band 8 (NIR), Band 12 (SWIR)
Dynamic World	10 meters	2019–2023	ESA	N/A

MCD64A1 data are stored in HDF (Hierarchical Data Format), containing multiple data layers, such as burned area masks, burn dates, and quality assurance information. These data can be accessed through NASA's Earthdata platform or the Google Earth Engine. Despite the global consistency and high accuracy of the MCD64A1 dataset, its spatial resolution of 500 m makes it difficult to accurately capture the detailed spatial distribution of fire events and may lead to commission and omission errors. Therefore, within the proposed framework of this study, MCD64A1 serves as coarse localization data to help determine the approximate location of burned areas to a larger extent.

2.1.2. Sentinel-2 Optical Imagery and Dynamic World Data

The Sentinel-2 satellite is equipped with 13 bands, covering spectral ranges from visible to near-infrared and shortwave infrared, including blue (490 nm), green (560 nm), red (665 nm), and multiple near-infrared and shortwave infrared bands. The spatial resolution of Sentinel-2 ranges from 10 m to 60 m, with visible and near-infrared bands offering a 10 m resolution, which allows for detailed observations of the Earth's surface. The combination of the two Sentinel-2 satellites provides a five-day revisit period for any location on Earth, which is crucial for monitoring grassland and forest fires (as shown in Table 2).

The Dynamic World dataset, jointly launched by Google Earth Engine and the World Resources Institute, is a real-time, high-resolution land cover classification dataset generated from Sentinel-2 data. This dataset employs a deep learning model to process Sentinel-2 imagery and generates land cover types in near real time, including nine major land cover classes: water, trees, grass, crops, built-up areas/artificial surfaces, bare ground, shrubs, wetlands, and snow/ice. Dynamic World's spatial resolution is 10 m, making it highly valuable for dynamically monitoring surface changes caused by fires.

In this study, the Dynamic World dataset is used to extract grassland and forest categories. Specific steps include (1) land cover category selection: filtering and extracting "forest" and "grassland" categories based on land cover codes and generating binary masks for grassland and forest regions and (2) mask application: applying the generated masks to Sentinel-2 multispectral imagery to extract optical imagery data containing only grassland and forest regions. By combining the high-resolution optical data of Sentinel-2 and the land cover classification data of Dynamic World, grassland and forest areas can be accurately delineated, thereby enhancing the accuracy of fire monitoring.

2.1.3. Sentinel-2 Burned Area Candidates

The high spatial resolution (10 m) and multispectral information of Sentinel-2 provide more detailed surface information in fire monitoring compared to MODIS fire point data. However, Sentinel-2's high resolution and large data volume present challenges in data

processing and analysis. To effectively integrate these two complementary datasets, we developed a comprehensive multi-source data fusion framework within the SAFE method. This framework is designed to harness the high temporal resolution of MODIS MCD64A1 fire detections and the high spatial resolution of Sentinel-2 imagery to optimize burned area extraction. To fully utilize the advantages of Sentinel-2 data in combination with MODIS fire detection data, we designed a module to generate candidate burned areas using Sentinel-2 data. This module encompasses several critical steps:

First, the geographic location and occurrence date of fire points were obtained using MODIS fire point data, and Sentinel-2 images corresponding to the dates closest to the fire occurrence were extracted to capture post-fire surface changes. Specifically, we implemented a temporal alignment process where Sentinel-2 images were selected based on a $\pm$2-day window surrounding each MODIS-detected fire event. This temporal proximity ensures that the Sentinel-2 imagery reflects the immediate aftermath of the wildfire, which is essential for accurate burned area segmentation. Additionally, Sentinel-2 images from before the fire occurrence were extracted to meet the requirements of bitemporal change detection. This pre-fire imagery serves as a baseline for identifying and quantifying the extent of burned areas by comparing changes in spectral reflectance and vegetation indices.

To ensure that Sentinel-2 imagery effectively covers all fire point areas, we performed a spatial topology analysis on the fire points detected by MODIS, merging spatially adjacent or nearby fire points into larger contiguous areas and calculating their bounding boxes to determine the extent of Sentinel-2 imagery to be clipped. This spatial aggregation minimizes redundancy and optimizes the area of interest for high-resolution analysis. Furthermore, we employed advanced spatial clustering algorithms, such as DBSCAN (Density-Based Spatial Clustering of Applications with Noise), to accurately group fire points based on their spatial proximity and occurrence patterns. This approach enhances the precision of burned area delineation by ensuring that all relevant fire points are cohesively analyzed within their geographic context.

To prevent omission, especially in marginal or sparsely distributed fire point areas, a 5 km spatial extension (equivalent to 20 MODIS pixels) was applied to the bounding box to ensure that potentially omitted fire areas in the fire point product were fully covered. This spatial buffer accounts for the inherent uncertainties and positional inaccuracies in MODIS fire detections, ensuring comprehensive coverage of all affected regions. Additionally, we integrated land cover information from the Dynamic World dataset to refine the spatial boundaries of burned areas, distinguishing between actual burned regions and non-burned land cover types that may exhibit similar spectral characteristics. This process generated candidate burned areas while excluding large amounts of non-burned areas, thereby enhancing the efficiency and accuracy of subsequent model operations.

2.1.4. Generation of Burned Area Candidate Boxes

Refining MODIS-scale burned areas to the Sentinel-2 scale is a crucial part of the methodology proposed in this study. This process is achieved by verifying the quality of the fire point product data, combined with the Burned Area Index BAIS2 calculated from Sentinel-2 imagery.

First, the BAIS2 index is calculated using the spectral characteristics of Sentinel-2 to evaluate the accuracy of the fire point product data. When the fire point data align with the BAIS2 index, it indicates that the fire point data in the area have undergone rigorous validation and are highly reliable. This means that the fire point data can be used as input for subsequent modules to extract burned areas at the Sentinel-2 scale comprehensively. Through this process, the accuracy and reliability of the data used for further processing are ensured, laying a solid foundation for precise burned area identification. By effectively

combining the infrared and visible bands of Sentinel-2, the BAIS2 index is better suited for refining burned area identification at the Sentinel-2 scale compared to other burned area indices, such as the Burned Area Index (BAI) and the Normalized Burn Ratio (NBR). Furthermore, if discrepancies are found between the fire point data and the BAIS2 results, a histogram analysis is used to further investigate abnormal areas.

BAIS2 is a fire detection index specifically designed for the spectral characteristics of Sentinel-2, with its innovation lying in the combination of the red-edge band ratio (used to describe vegetation characteristics) and the response of the shortwave infrared (SWIR) band. This combination shows significant efficacy in determining burned areas, enabling the more precise detection and identification of fire-affected regions. Specifically, when the fire point data align with the BAIS2 results, it suggests that the data are reliable and can be used for the subsequent comprehensive extraction of burned areas. This validation process ensures the data quality, laying a solid foundation for further analysis. The *BAIS*2 formula is as follows:

$$BAIS2 = (1 - \sqrt{\frac{B6 \times B7 \times B8A}{B4}}) \times (\frac{B12 - B8A}{\sqrt{B12 + B8A}} + 1) \quad (1)$$

where $B4$, $B6$, $B7$, $B8A$, and $B12$ correspond to the Red, Red Edge 2, Red Edge 3, NIR Narrow, and SWIR2 bands of Sentinel-2, respectively.

2.2. High-Quality Burned Area Data

Through the aforementioned steps, preliminary screening of MODIS MCD64A1 fire points and Sentinel-2 optical imagery was conducted. The next step involves the second phase of burned area generation, in which the spatial distribution of fire point product data is refined using Sentinel-2 imagery to generate high-quality burned area data.

2.2.1. Segment Anything Model (SAM)

To efficiently extract burned areas from higher-resolution Sentinel-2 imagery while avoiding the computational and time costs associated with local training, we employed the SAM model, a large-scale semantic segmentation model. SAM, initially proposed in the field of natural image segmentation, is known for its excellent segmentation performance and strong generalization capabilities, allowing for zero-shot transfer across different scenarios. The SAM model is composed of three primary modules (Figure 2): image encoding, prompt encoding, and mask decoding. The image segmentation process proceeds as follows: first, SAM uses a pre-trained Vision Transformer (ViT) to process the input image, generating an embedded representation of the image during the encoding process. This embedding serves as a compressed and abstract representation of the input image, providing foundational information for subsequent mask generation. Prompt encoding is another critical component of the SAM model, capable of processing various prompt types, including simple spatial cues (such as points or bounding boxes) and more complex mask prompts (such as textual inputs). These prompts are converted into a format understandable by SAM, guiding it to identify and segment specific objects in the image. Finally, the mask decoder combines the image embedding and prompt encoding to generate the final segmentation mask. For ambiguous prompts that may refer to multiple objects, SAM can generate multiple masks, each with its own confidence score, which allows it not only to identify objects in the image but also to assess the accuracy of the predictions:

$$Seg_{img} = V_{mask}(U_{img}(x), U_{pro}(p_{point})) \quad (2)$$

where U_{img} denotes the image encoder, U_{pro} corresponds to the prompt encoder, and V_{mask} refers to the mask decoder. x denotes a remote sensing image waiting to be pro-

cessed and P_{point} is the prompt information inputted by users. Seg_{img} corresponds to the predicted masks.

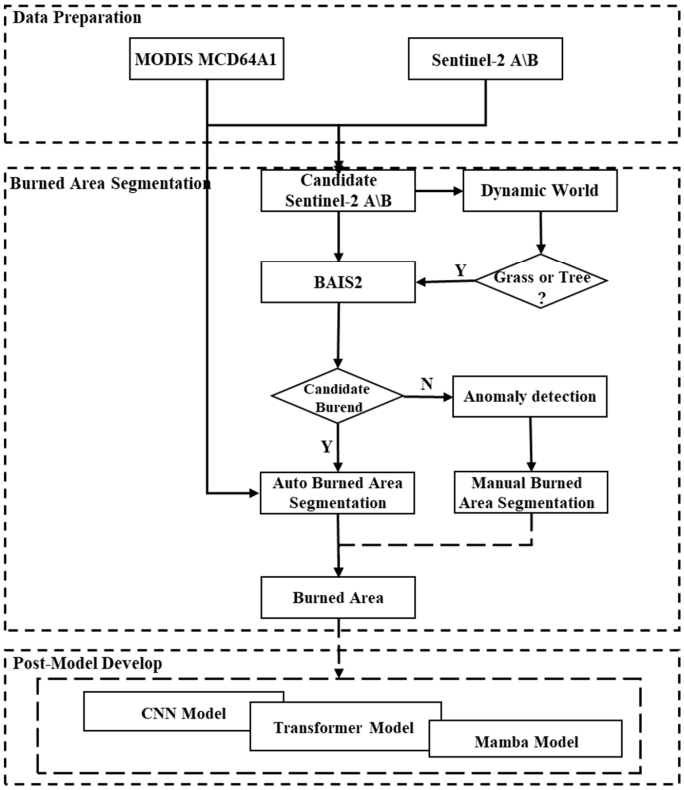


Figure 2. Flowchart of the SAFE model.

The use of user-defined prompts bridges the gap between natural image segmentation and user-defined segmentation tasks, distinguishing SAM from traditional semantic segmentation models. SAM uses prompts to directly guide the model to specify regions or objects to be segmented (as shown in Figure 3). This flexibility enables users to dynamically adjust the segmentation results by modifying the prompts, thereby enhancing the model's applicability and adaptability to diverse applications.

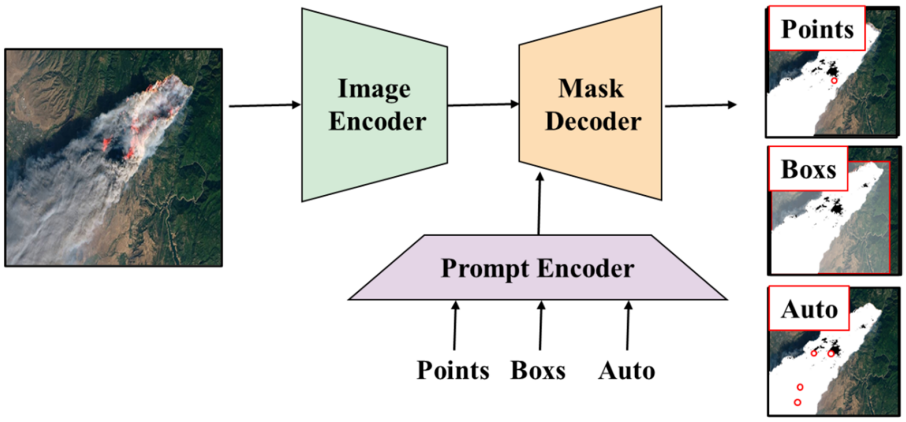


Figure 3. Framework of the Segmentation Anything Model (SAM). Red dots represent input SAM prompt points, and red boxes represent input SAM prompt boxes.

2.2.2. High-Resolution Burned Area Extraction Module: SAFE

Based on the calibrated distribution of prompt points, the SAFE module further extracts high-resolution burned areas (as shown in Figure 4), following these steps:

(1) Generating bounding boxes and inputting to SAM: First, bounding boxes are generated based on the product data from the initial step, utilizing the spatial distribution of fire points. These bounding boxes cover potential fire-affected regions and serve as input prompts for the SAM model to precisely indicate the areas to be segmented. The generation of bounding boxes considers both the spatial locations and shape characteristics of the fire points to ensure that all possible burned areas are encompassed.

(2) Segmentation results generation by SAM: SAM uses the input bounding boxes to segment the image, generating the distribution of burned areas within that region. Specifically, SAM first processes the input image using the image encoding module, generating an embedded representation that captures the main features of the input image. It then encodes the input prompts through the prompt encoding module, transforming the bounding box information into a format that the model can interpret. The combination of image embedding and prompt encoding allows SAM to more accurately identify burned areas.

(3) Mask decoding and generation: The mask decoder of SAM integrates the image embedding and prompt encoding to generate masks for the corresponding burned areas. The mask decoder is a critical component, converting high-dimensional embeddings into concrete mask images that reveal the target regions segmented from the image. When prompts potentially refer to multiple objects, SAM is capable of generating multiple masks, each with a corresponding confidence score, thereby assessing the segmentation accuracy. This ability to generate multiple masks makes SAM highly flexible and accurate when dealing with complex scenes.

(4) Generating high-resolution burned area result maps: Finally, the segmentation masks produced by SAM are used to create high-resolution burned area result maps. These result maps provide precise depictions of the burned areas within the bounding box regions, significantly enhancing the accuracy of burned area identification. During the result generation process, the system also applies post-processing to each mask to eliminate noise and artifacts, ensuring the reliability and clarity of the results. Through this process, the SAFE module generates high-resolution burned area maps, providing critical data support for subsequent fire impact assessments and emergency responses.

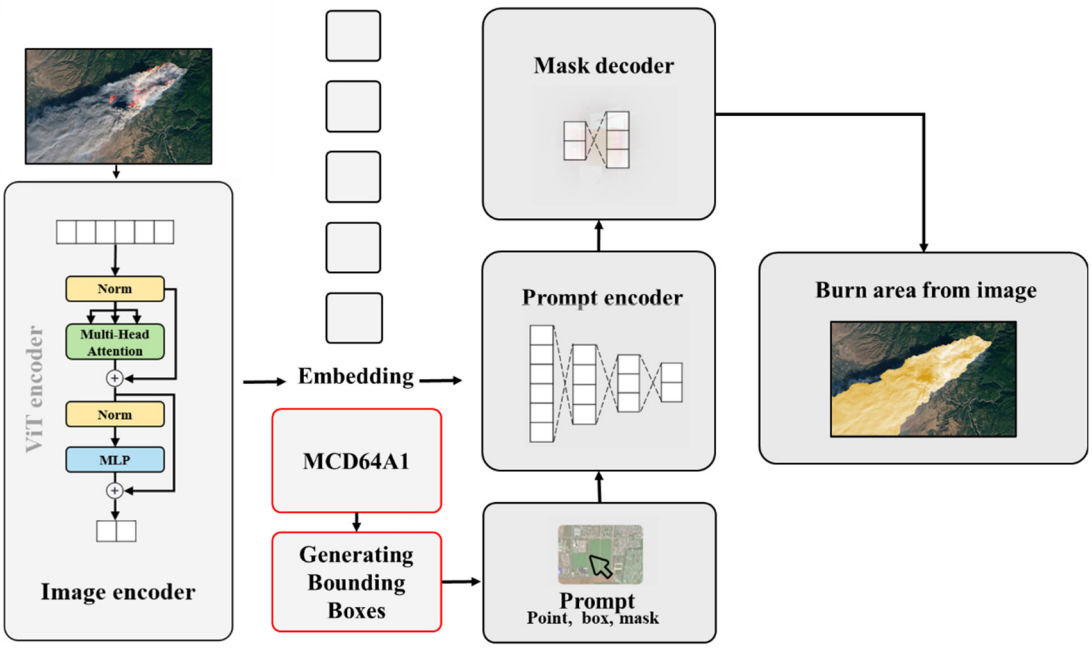


Figure 4. Schematic diagram of the SAFE model.

2.2.3. Supplementing Burned Area Samples

In some regions, the automatically generated burned area data may be insufficient, particularly in areas with complex terrain or diverse land cover types. The automatic extraction results in these regions are often affected by factors such as ambiguous remote sensing image features, complex vegetation types, and cloud cover, leading to inaccuracies in burned area identification and significant discrepancies from the actual burned area extent. To further enhance the accuracy and completeness of the burned area data, we propose a semi-supervised human-in-the-loop annotation strategy to address the shortcomings of automatic extraction.

First, for product data that did not pass quality checks, related information can be extracted from Sentinel-2 imagery. These high-resolution and multispectral images provide more detail for identifying burned areas. Based on this information, human annotators can supplement these images using a semi-supervised approach to ensure the comprehensive and precise labeling of burned areas, particularly in regions where automatic detection is challenging.

To further reduce manual workload in this process, we designed and introduced an extraction tool based on the Segment Anything Model (SAM). SAM is a powerful segmentation tool that supports various interaction methods, including point and box selection, allowing users to annotate images more efficiently. Specifically, users can quickly annotate target areas by selecting points or drawing boxes on Sentinel-2 imagery. During this process, SAM can automatically infer the boundaries of regions of interest, combining human supervision and correction to maintain high efficiency while ensuring annotation accuracy (https://github.com/LiuSjun/SAFE (accessed on 15 December 2024)).

This SAM-based annotation tool not only supplements the current dataset but also provides an efficient solution for future fire monitoring and identification in areas with complex terrain and diverse land cover. Through this human–machine collaboration approach, the precision of burned area extraction and the comprehensiveness of data coverage are significantly improved.

2.3. Model Implementation: Local Model Training

Supervised deep learning models play a key role in burned area extraction. By learning the features of training data, these models can delve deeper into capturing the spatial, spectral, and spatiotemporal characteristics of local burned areas, thereby further improving the accuracy and completeness of burned area extraction within a given region. In practical applications, these models not only accurately identify post-fire vegetation changes based on satellite imagery but also effectively distinguish burned areas from non-fire regions with similar spectral characteristics, thereby enhancing the reliability and scientific validity of the extraction results.

A significant challenge in extracting burned areas is the considerable variability between different geographic regions. Due to factors such as terrain, vegetation type, and climatic conditions, the mechanisms and manifestations of fires vary greatly from one region to another, which can limit the effectiveness of a generalized fire extraction model across different regions. Furthermore, fire-prone areas are often concentrated in specific regions, while fire occurrences are relatively sparse at the global or national scale. Consequently, directly designing and training a high-resolution fire extraction model across a large area would result in substantial computational resource wastage, particularly given the high demands for high-performance computing equipment and data storage, rendering such an approach inefficient and costly.

To address these challenges, a region-specific, targeted training strategy can be adopted. First, high-resolution burned area distribution maps for each region are generated through

preliminary processing and existing fire records, which serve as reference label data for subsequent model training. Specifically, we compiled a dataset comprising 15,000 Sentinel-2 image patches, each annotated with precise burned area labels derived from MODIS MCD64A1 fire detections and validated using high-resolution aerial imagery. This dataset was divided into 12,000 training samples, 2000 validation samples, and 1000 test samples to ensure robust model evaluation. Then, state-of-the-art (SOTA) deep learning models, such as Transformer, ViT, and Mamba [58,73,74], can be utilized to adjust and train models for each specific study region. For each target model, we employed a fine-tuning approach using transfer learning, initializing the models with weights pre-trained on large-scale image datasets like ImageNet to enhance their feature extraction capabilities in remote sensing contexts.

In this study, we conducted localized model training for specific regions using supplementary sample data. The training process involved optimizing hyperparameters including a learning rate of 1×10^{-4}, a batch size of 32, and training for 100 epochs with early stopping based on validation loss to prevent overfitting. We utilized the Adam optimizer with default $\beta 1$ and $\beta 2$ values and implemented a combination of Cross-Entropy Loss and Dice Loss to address the class imbalance and improve the segmentation performance. By incorporating more samples relevant to the target region, the model's adaptability and accuracy in the local environment were enhanced, effectively improving its ability to capture fire characteristics. Additionally, data augmentation techniques such as random rotations, horizontal and vertical flips, and scaling were applied to increase the diversity of the training data, thereby enhancing the model's generalization capabilities. This localized training strategy enables the model to achieve higher accuracy and reliability when addressing fire extraction tasks in different regions, providing more precise tools for fire monitoring and post-fire assessment.

3. Experiments

3.1. Testing Areas

To evaluate the accuracy of the proposed burned area extraction algorithm, this study collected high-resolution remote sensing images from various regions of the world, covering different time periods after fire events (as shown in Figure 5). The primary data used were Sentinel-2 multispectral images. These data provide a high temporal resolution with 10 m imagery, which is used to map burned area distributions and validate accuracy based on spectral imagery. The selected areas, as shown in the table, cover different continents and diverse geographical environments, providing robust conditions for algorithm validation.

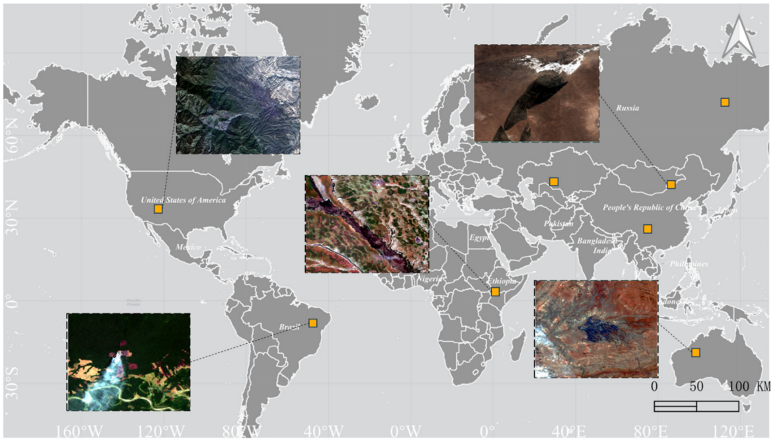


Figure 5. Basic information of the study area: spatial distribution of image acquisition locations.

Based on the differences in the selected scenarios, we classified the study areas into two major types for validation and discussion: grassland fire and forest fire scenarios, to verify the feasibility of model extraction. In addition, we also selected a larger area at the border between China and Mongolia for year-round grassland fire testing.

3.2. Datasets

Sentinel-2, developed and operated by the European Space Agency (ESA), provides free optical data with high temporal and spatial resolution and rich spectral information to support global environmental and security monitoring [75]. Two Sentinel-2 products are hosted in Google Earth Engine (GEE): Level-1C Top-of-Atmosphere (TOA) reflectance and Level-2A surface reflectance (SR) images, with the latter obtained by performing atmospheric correction on the Level-1C product. In this study, we used Sentinel-2 Level-2A SR imagery hosted in GEE, primarily because it includes correction for terrain-induced distortions, such as mountain shadows [76], which to some extent overcomes the negative impact of shadows on extraction accuracy in mountainous areas.

To evaluate the robustness and feasibility of the proposed BA algorithm, we collected 23,214 samples from different regions worldwide between June 2018 and October 2022. We used the Labelme tool to manually label these burned areas, referring to pre-fire imagery to ensure annotation accuracy, and generated burned area labels for these regions.

3.3. Evaluation Indices

We used four metrics to evaluate the performance of BA detection: overall accuracy (OA), precision, Kappa coefficient (KC), and F1-score. These metrics are commonly used to assess the quality of burned area mapping. Definitions of each metric are as follows:

Overall accuracy (*OA*): the OA quantifies the proportion of correctly predicted samples among all samples, calculated as follows:

$$OA = \frac{TP + TN}{TP + TN + FP + FN} \tag{3}$$

where *TP* (True Positive) represents correctly predicted positive samples, *TN* (True Negative) represents correctly predicted negative samples, *FP* (False Positive) represents negative samples incorrectly predicted as positive, and *FN* (False Negative) represents positive samples incorrectly predicted as negative.

Kappa coefficient (*KC*): the Kappa coefficient evaluates the consistency between predicted and actual classifications, accounting for agreement by chance, calculated as follows:

$$KC = \frac{p_0 - p_e}{1 - p_e} \tag{4}$$

where p_0 represents the observed agreement, i.e., the proportion of correctly classified samples, and p_e represents the expected agreement by chance, i.e., the probability of correct prediction under random conditions.

*F*1-score: F1-score is the harmonic mean of precision and recall, providing a balance between them, calculated as follows:

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{5}$$

where *precision* is the proportion of correctly predicted positive samples among all predicted positive samples and *recall* is the proportion of correctly predicted positive samples among all actual positive samples.

Higher values for the *OA*, *F*1-score, and *KC* indicate better extraction accuracy, while higher values for the *F*1-score indicate better performance [77,78].

3.4. Experiments

Although the proposed method can be directly applied without training, training data were still used for comparative deep learning methods. Therefore, we split the sample data, randomly selecting a total of 17,410 samples (75% of the total) from different regions as training data. After 100 training epochs, the model with the minimum loss and highest accuracy in each continent was selected to test the performance of comparative methods on validation datasets from each continent.

To evaluate the effectiveness of the proposed model, we compared our method with traditional BAIS2 index methods as well as the latest deep learning models U-Net and BiAU-Net. This comparative analysis allows us to verify the superiority of our proposed method over traditional algorithms and current state-of-the-art supervised deep learning approaches. To this end, we used the aforementioned four evaluation metrics to analyze the mapping results, providing valuable insights into the effectiveness of the model.

3.4.1. Experiment I: Burned Area Detection in Grassland Regions

We conducted experiments for burned area extraction in grassland regions. The grassland fire extraction experiment aimed to monitor the occurrence of grassland fires and the burned areas after fires using remote sensing technology. High-resolution Sentinel-2 remote sensing data were used in the experiment, collecting post-fire imagery for burned area extraction. Since grassland vegetation recovers quickly, the focus of this experiment was on capturing short-term vegetation changes.

The selected test sites included Khorgoljyn in southern Mongolia, Aktobe in northwestern Kazakhstan, Coober Pedy in southern Australia, and Monduli in northeastern Tanzania (as shown in Figure 6). The data for these sites included grassland fire scenes from October 2018 to May 2022. During the experiment, Sentinel-2 data were first preprocessed, including radiometric correction, atmospheric correction, and geometric correction, to ensure data consistency and high quality. Subsequently, traditional threshold-based BAIS2 (Burned Area Index) was used to extract burned areas, and the results were compared with deep learning algorithms U-Net, BiAU-Net, and SAFE. To verify the extraction accuracy, existing fire distribution maps were used as a reference for comparative evaluation.

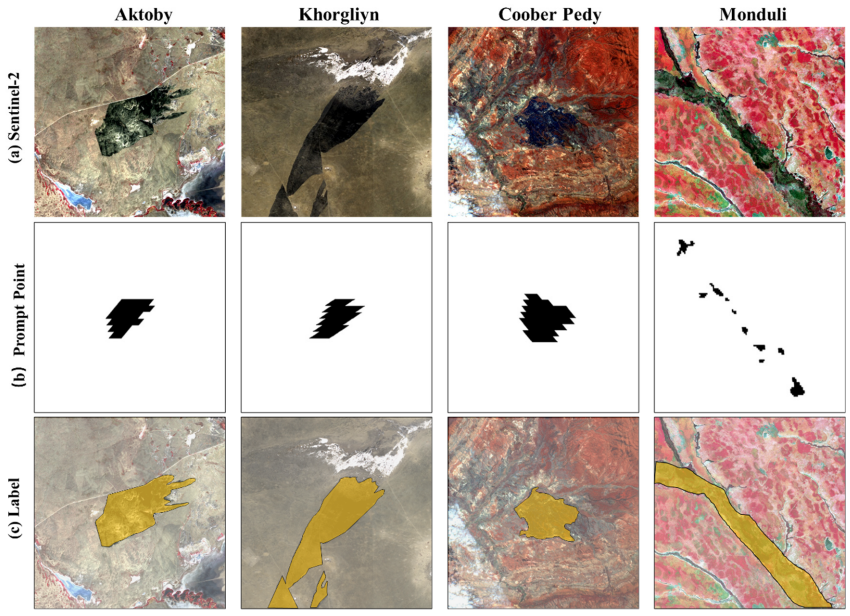


Figure 6. Burned area Sentinel-2 images of grassland regions (**a**), MCD64A1 generated prompt points (**b**), and labels created with Labelme (**c**).

3.4.2. Experiment II: Burned Area Detection in Forest Regions

To detect burned areas in forest regions, we designed an experiment to monitor forest fires and extract burned areas after fires using remote sensing technology. This experiment also used high-resolution Sentinel-2 remote sensing data, collecting post-fire imagery to monitor and analyze burned areas. Since forest vegetation recovers slowly, the experiment focused on the long-term process of post-fire vegetation changes and a fire impact assessment.

The experiment selected fire-prone areas, including the Mendocino National Forest in North America, Parque in the Amazon rainforest of South America, Xichang in southwestern China, and Yakutsk in Russia (as shown in Figure 7). These areas are representative, covering different types of vegetation, terrain, and climatic conditions, to test the applicability of the algorithm in diverse scenarios.

The data used in the experiment included multiple forest fire scenes from 2019 to 2023. Traditional threshold-based BAIS2 was used for burned area extraction, and the results were compared with those of deep learning algorithms U-Net, BiAU-Net, and SAFE.

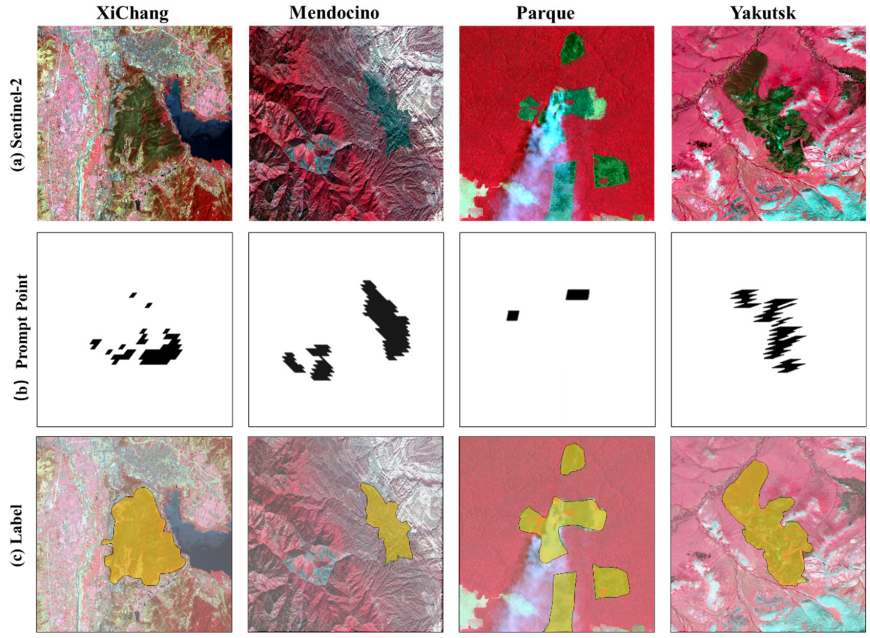


Figure 7. Burned area Sentinel-2 images of forest regions (**a**), MCD64A1 generated prompt points (**b**), and labels created with Labelme (**c**).

3.4.3. Experiment III: Application Example of Burned Area Detection

To evaluate the applicability and effectiveness of the designed burned area extraction algorithm at a large scale, we conducted an experiment for large-scale application testing. This experiment aimed to examine the performance of the burned area extraction algorithm in different geographic regions and ecological environments to verify its generalization ability and robustness. The experiment selected the Inner Mongolia grasslands in northeastern China as the test site. This region is frequently affected by fires and has abundant ground observation data to support the experiment, providing a reliable reference. The dataset selected for the experiment covered a topological area of 50 km around the border between Inner Mongolia and Mongolia from 2020 to 2021, covering millions of hectares (as shown in Figure 8). The standardized preprocessing of Sentinel-2 data, including radiometric correction, atmospheric correction, and geometric correction, was conducted to ensure data consistency across different regions.

In the experiment, the designed burned area extraction algorithm was applied for the unsupervised extraction of burned areas across the region. To enhance accuracy, we

introduced localized Transformer model training based on extraction results generated by SAFE, allowing for a more refined adaptation to regional characteristics. To ensure reliability and objectivity, the burned areas extracted in each region were compared with publicly available fire monitoring databases. Evaluation metrics were then employed to comprehensively assess the algorithm's performance, including the improvement in accuracy achieved through localized model training for large-scale fire monitoring.

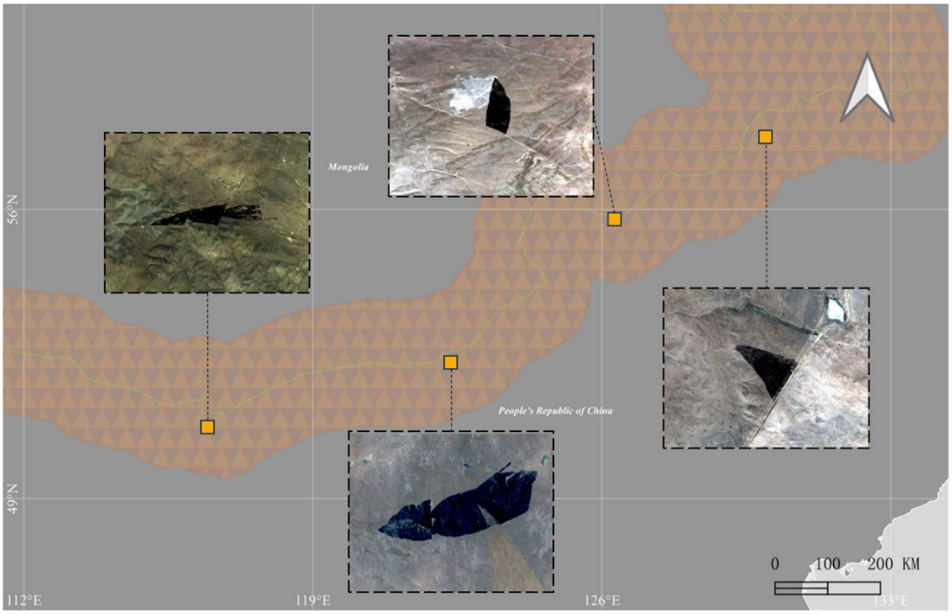


Figure 8. Burned areas within the China–Mongolia border region.

4. Results

4.1. Experiment I

As shown in Figure 9, which compares the Sentinel-2 reference image of burned areas in grassland (a) with the burned area extents extracted by the BAIS2 threshold method (b), U-Net (c), BiAU-Net (d), and SAFE (e), it is evident that SAFE demonstrated a superior performance in capturing the complexity of burned areas. Figure 10 provides a comparison of the F1-score, Kappa, and mIoU of burned area extraction in grassland regions by BAIS2, U-Net, BiAU-Net, and SAFE. In all test areas, the SAFE showed the best performance, particularly in the Aktoby and Khorgliyn regions, where the F1-scores reached 0.89 and 0.88, Kappa values were 0.86 and 0.85, and mIoU values were 0.80 and 0.79, respectively. These results highlight SAFE's ability to accurately delineate complex burned area boundaries through the effective integration of multi-source detection outputs.

This demonstrates that SAFE is better at capturing the details of burned areas, especially in complex grassland environments (as shown in Figure 10). U-Net and BiAU-Net also performed well, with the F1-score, Kappa, and mIoU metrics significantly higher than those of the BAIS2 method across all regions, indicating the consistency, robustness, and accuracy in the boundary extraction of deep learning algorithms for burned area extraction. For example, in the Monduli region, the F1-scores for U-Net and BiAU-Net were 0.83 and 0.84, Kappa values were 0.80 and 0.81, and mIoU values were 0.75 and 0.76, significantly outperforming the BAIS2 method.

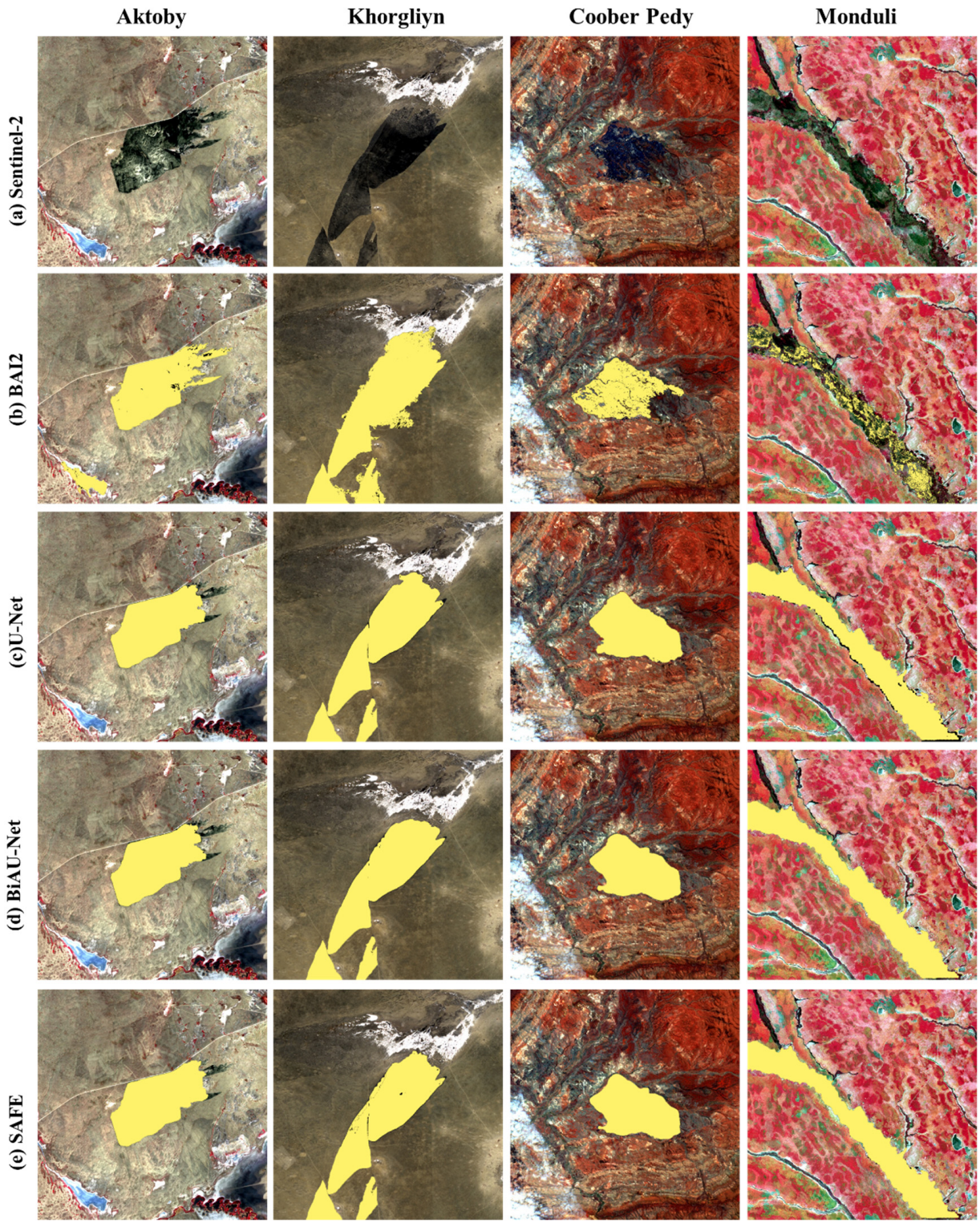


Figure 9. Sentinel-2 reference image of burned areas in grassland (**a**), burned area extent extracted by BAIS2 threshold method (**b**), U-Net (**c**), BiAU-Net (**d**), and SAFE (**e**).

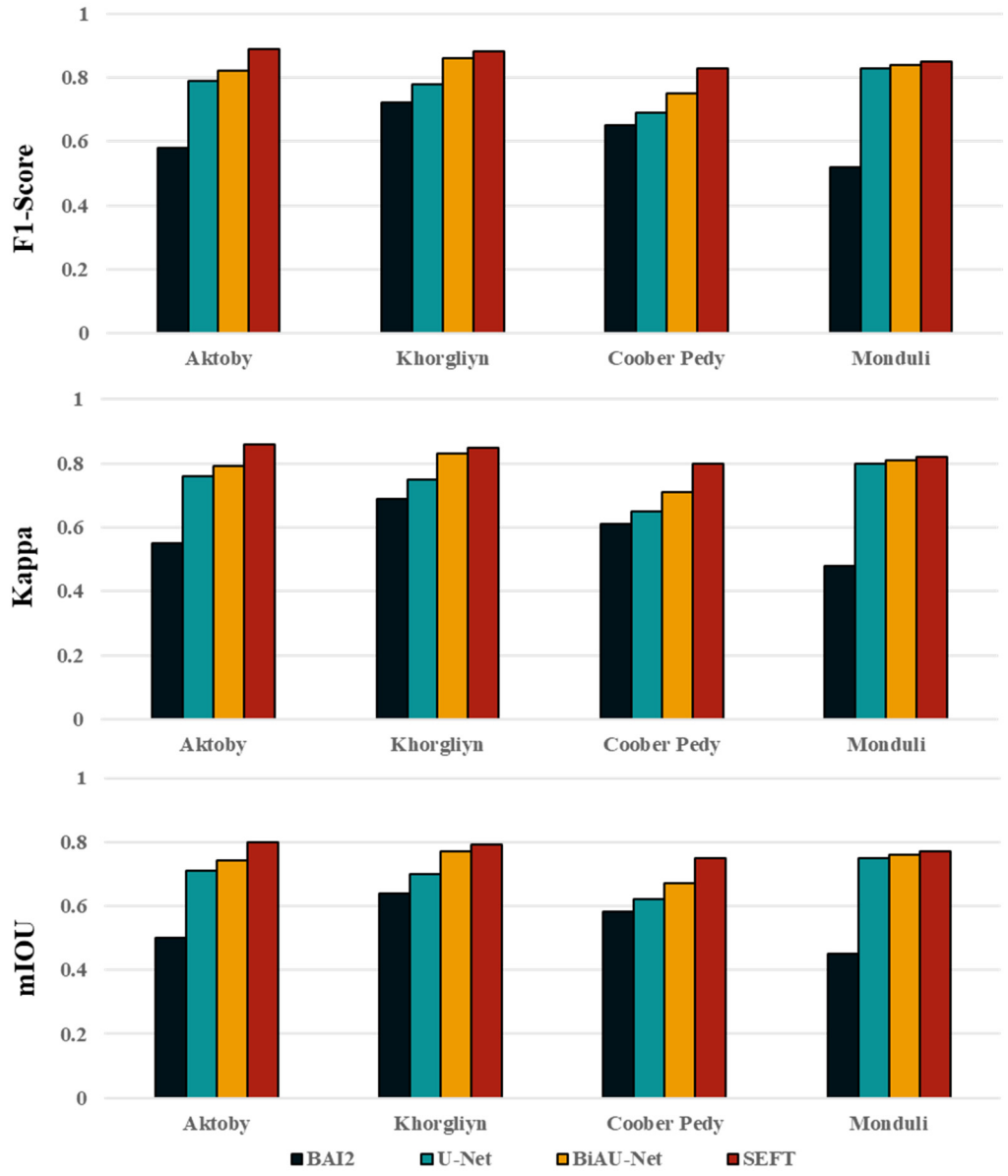


Figure 10. F1-score, Kappa, and mIOU of burned area extraction in grassland regions by BAIS2, U-Net, BiAU-Net, and SAFE.

4.2. Experiment II

This experiment evaluated the performance of the traditional threshold method BAIS2 and deep-learning-based algorithms U-Net and BiAU-Net. The results were comprehensively compared, revealing the performance of each algorithm in different forest fire scenarios. Overall, the deep-learning-based methods were superior to the traditional BAIS2 method (as shown in Figure 11). In all test areas, the SAFE showed the best performance, particularly in the Parque and Mendocino regions, where the F1-scores reached 0.89 and 0.84, Kappa values were 0.85 and 0.80, and mIoU values were 0.80 and 0.76, respectively (as shown in Figure 12), representing a significant improvement over the BAIS2 method. This indicates that deep learning algorithms can better capture the details of burned areas, especially in complex forest environments.

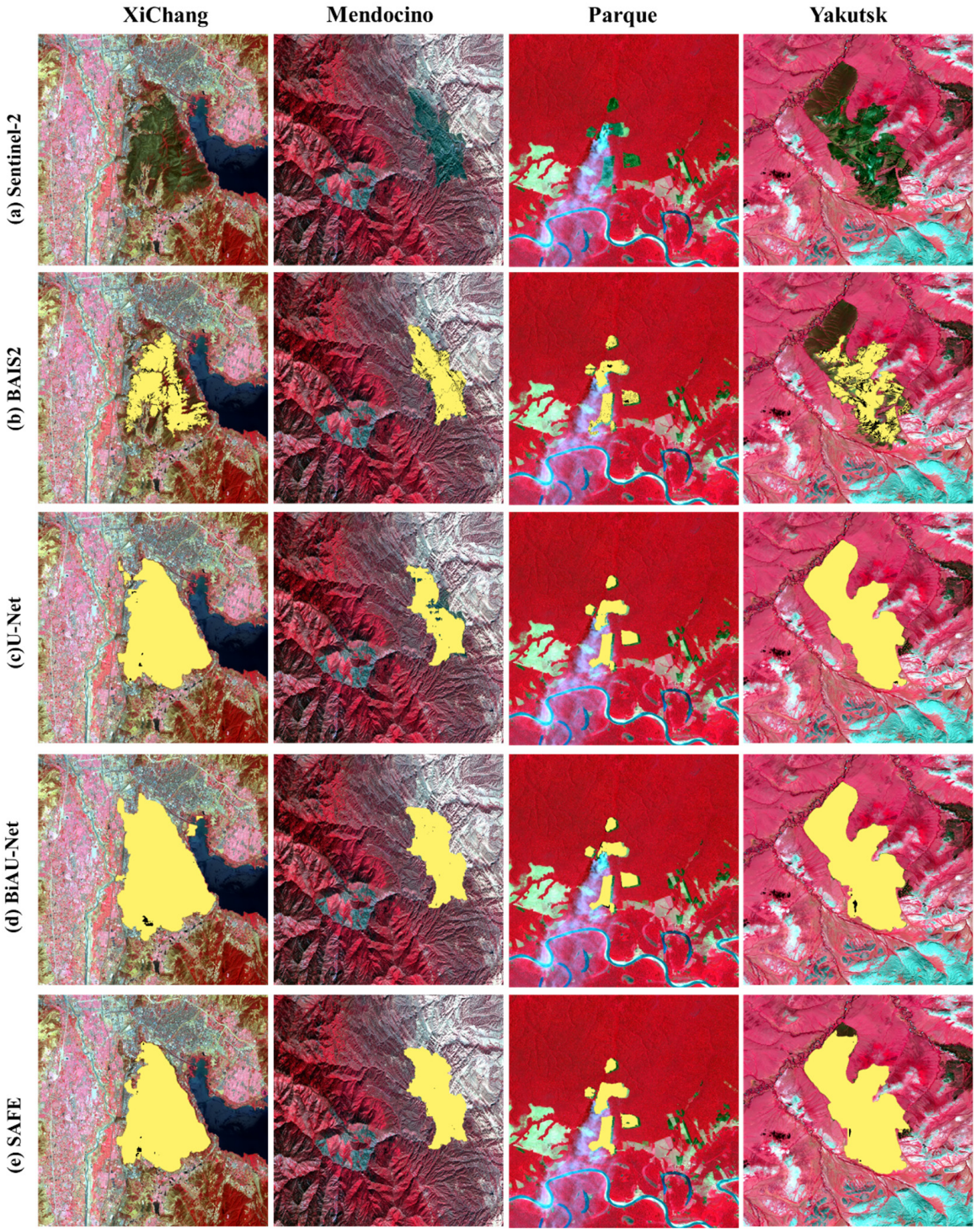


Figure 11. Sentinel-2 reference image of burned areas in forest (**a**), burned area extent extracted by BAIS2 threshold method (**b**), U-Net (**c**), BiAU-Net (**d**), and SAFE (**e**).

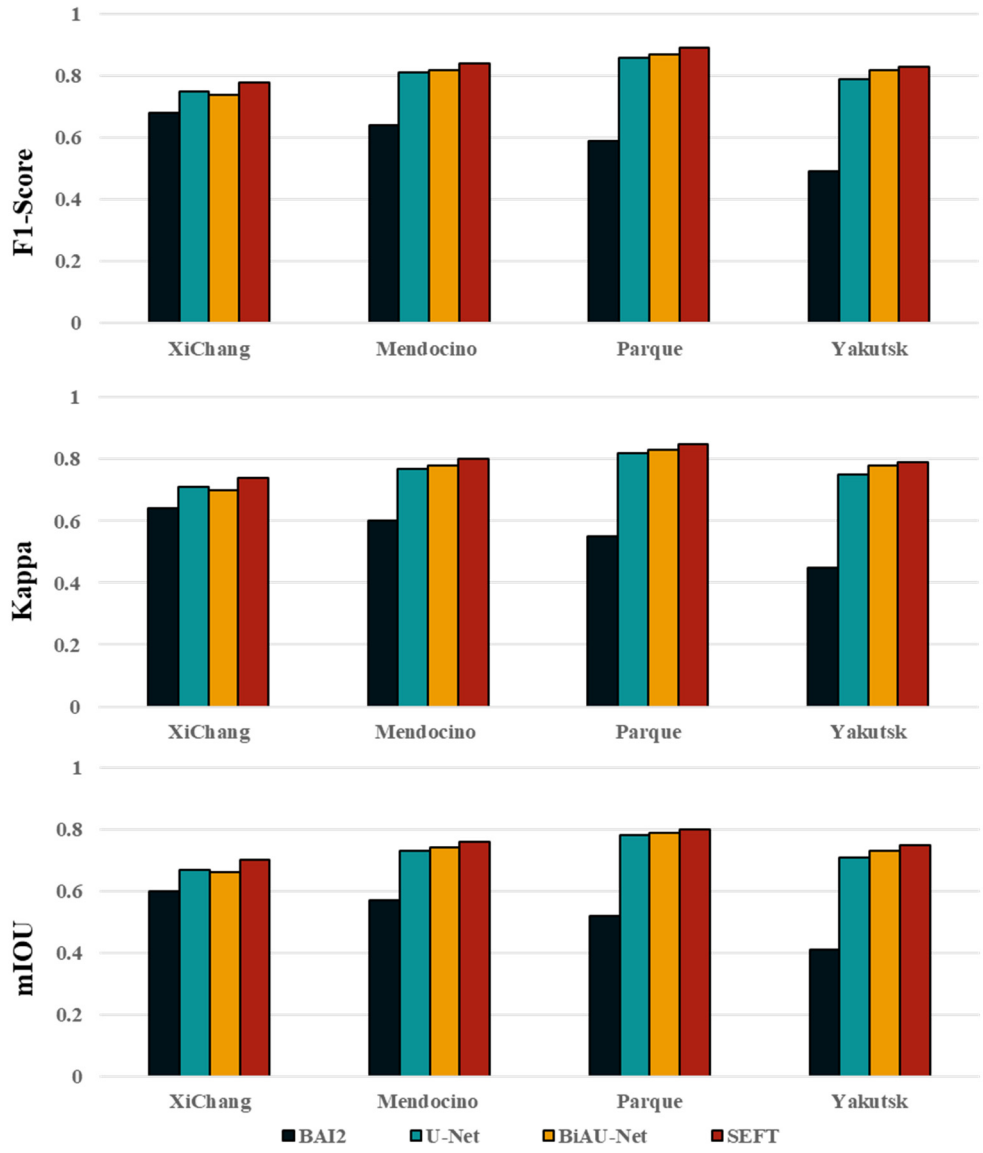


Figure 12. F1-score, Kappa, and mIOU of burned area extraction in forest regions by BAIS2, U-Net, BiAU-Net, and SAFE.

U-Net and BiAU-Net also performed well, with the F1-score, Kappa, and mIoU metrics significantly higher than those of the BAIS2 method across all regions, demonstrating the consistency, robustness, and accuracy in the boundary extraction of deep learning algorithms for burned area extraction. For instance, in the Xichang and Yakutsk regions, the F1-scores for U-Net and BiAU-Net were 0.75 and 0.79, Kappa values were 0.71 and 0.75, and mIoU values were 0.67 and 0.73, significantly outperforming the BAIS2 method.

The experimental results indicate that deep-learning-based burned area extraction algorithms perform significantly better than traditional BAIS2 methods in forest fire scenarios. The SAFE, in particular, achieved the best results across all evaluation metrics, demonstrating its ability to accurately extract burned areas in diverse forest environments.

4.3. Experiment III

This experiment conducted a large-scale application test in the Inner Mongolia grasslands of northeastern China to evaluate the applicability and effectiveness of the proposed burned area extraction algorithm. To better adapt to the specific characteristics of different regions, a locally trained Transformer model, using labels generated by SAFE, was

employed for enhanced burned area detection. The experimental results showed that this locally trained Transformer outperformed the traditional BAIS2 and SAFE method across all metrics, achieving the best results in accuracy and reliability (as shown in Figure 13).

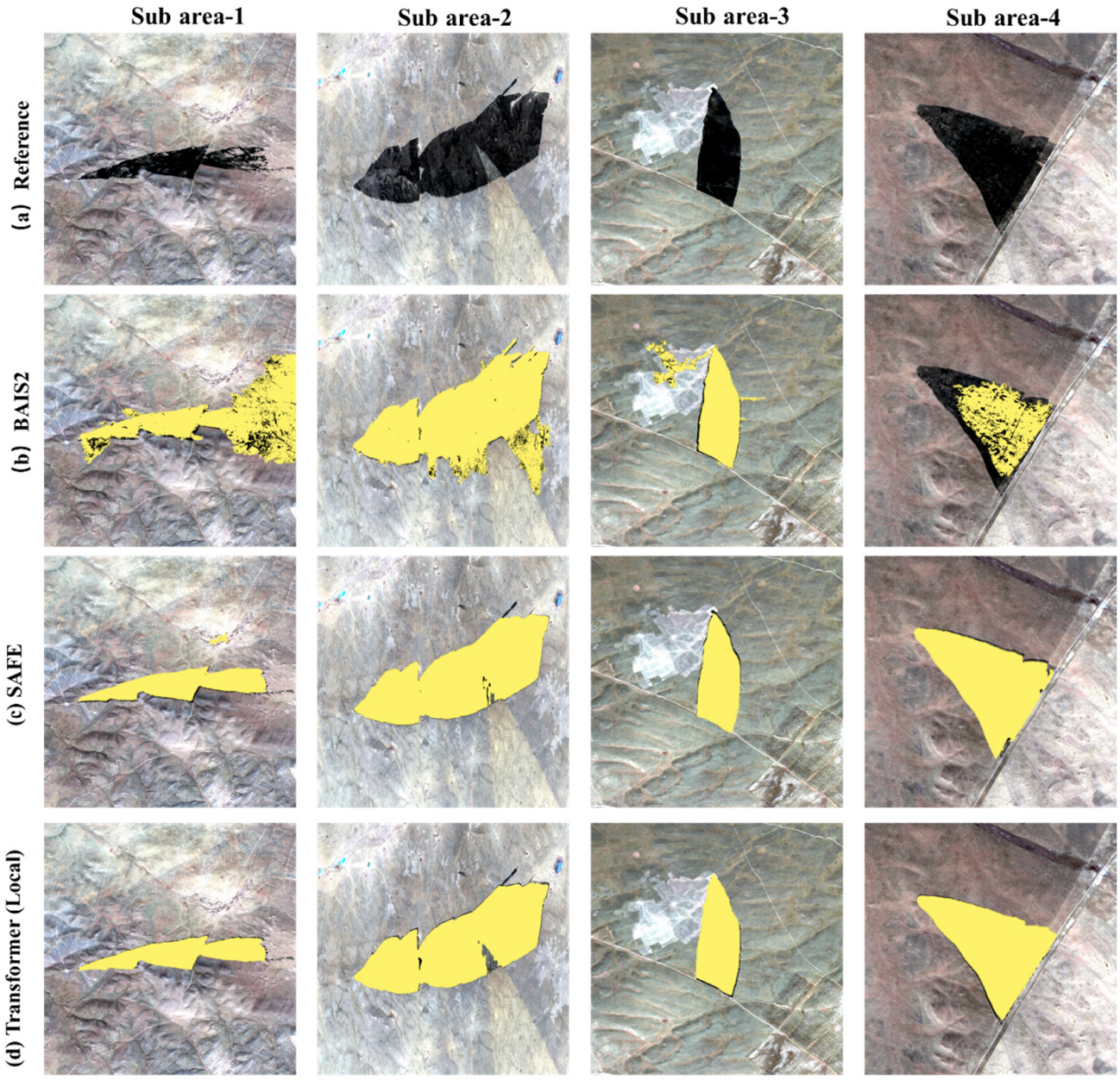


Figure 13. Sentinel-2 reference image of burned areas in the China–Mongolia border region (**a**), burned area extent extracted by BAIS2 threshold method (**b**), SAFE (**c**), and Local Transformer (**d**).

In all sub-regions, the F1-scores of the locally trained Transformer reached or exceeded 0.9, indicating a high level of accuracy in fire extraction. In contrast, the F1-score of the BAIS2 and SAFE methods ranged from 0.51 to 0.85, significantly lower than the deep learning algorithms (as shown in Figure 14), highlighting the limitations of traditional methods in complex grassland environments.

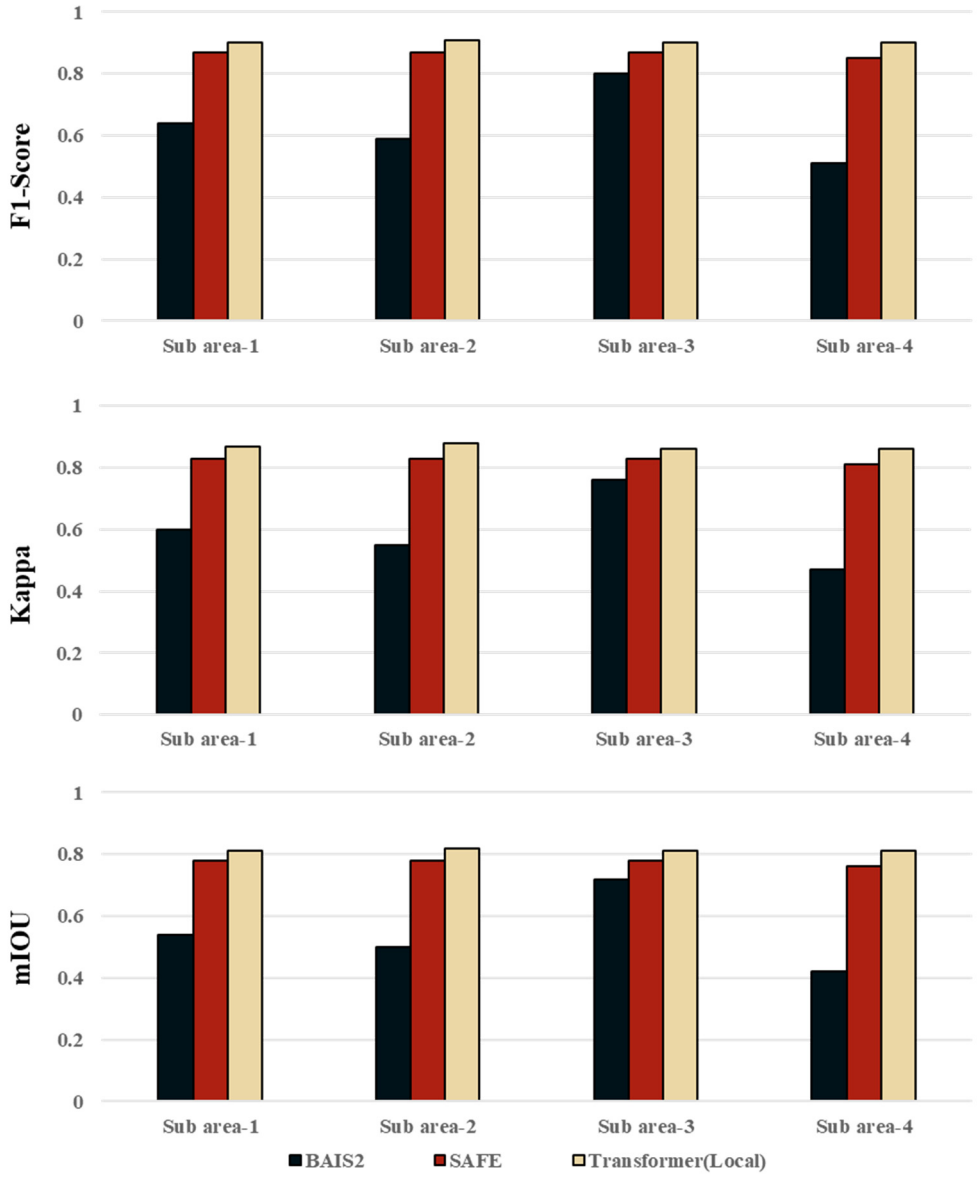


Figure 14. F1-score, Kappa, and mIOU of burned area extraction in forest regions by BAIS2, SAFE, and Transformer (Local).

For the Kappa coefficient, the locally trained Transformer achieved values of 0.88 and 0.86 in Sub-area 2 and Sub-area 4, respectively, showing excellent consistency and stability in burned area extraction, even in challenging environments. The mIoU results also validated the superiority of a locally trained Transformer. In all sub-regions, the mIoU of SAFE and BAIS2 ranged from 0.76 to 0.82. The transformer demonstrated a high performance in F1-score and Kappa and further excelled in precision and coverage for burned area boundary extraction, underscoring its generalizability and robustness across complex geographic environments.

5. Discussion

The SAFE algorithm proposed in this study has demonstrated significant advantages in burned area extraction. Compared with traditional threshold methods like BAIS2 and deep learning models such as U-Net and BiAU-Net, SAFE has shown its efficiency and accuracy in unsupervised extraction. Experimental results indicate that SAFE outperformed BAIS2 in all test regions, with significantly higher F1-score, Kappa, and mIoU metrics, especially in complex grassland and forest environments where SAFE excelled in capturing

the details of burned areas. Compared with deep learning models, SAFE eliminates the training phase and directly extracts burned areas, significantly reducing the total extraction time while providing segmentation results comparable to or even better than those of supervised learning methods.

The main advantages of SAFE are as follows: SAFE can execute burned area extraction tasks at a high speed with high segmentation accuracy. For near-real-time burned area extraction tasks, SAFE can respond quickly while ensuring high resolution and detail completeness, making it suitable for fire monitoring tasks that require a timely response. In addition, SAFE inherits the strong robustness of the SAM model, which underwent large-scale and diversified training, allowing it to effectively handle burned area extraction tasks under various complex scenarios, including smoke occlusion, lighting variations, and different backgrounds. Especially when dealing with small and fragmented burned areas, SAFE can still perform effective segmentation, demonstrating strong generalization capabilities in complex geographical environments. Moreover, SAFE provides an unsupervised automated workflow for burned area extraction that can automatically segment all burned areas in an image while also supporting user interactions for more precise segmentation based on user input, thus improving extraction efficiency and enhancing application flexibility. This interactive feature of SAFE is particularly effective for tasks requiring specific adjustments based on user needs. The unsupervised nature of SAFE allows it to complete extraction tasks without the need for samples or training, which presents a significant advantage in terms of efficiency. Compared with the threshold-based BAIS2 method, which also does not require training, SAFE maintains a similar extraction time but achieves higher segmentation accuracy. Compared with deep learning models that require training, SAFE eliminates the need for a significant amount of training time (as shown in Table 3) while achieving more accurate extraction results, highlighting its considerable advantage in both extraction efficiency and accuracy. Furthermore, SAFE demonstrates excellent coverage and scalability, enabling the effective monitoring of large-scale areas. Its ability to process vast amounts of high-resolution satellite data ensures that extensive regions can be monitored simultaneously, facilitating comprehensive wildfire management and a timely response across diverse geographic landscapes. This scalability makes SAFE a valuable tool for national and global wildfire monitoring initiatives, where large-scale data processing and real-time analysis are crucial.

Table 3. The training and inference time of BAIS2, U-Net, BiAU-net, SAFE, and Transformer (Local) in experiment I, with inference time representing the time required to infer 100 test samples.

	Experiment II	
	Training Time (h)	**Inference Time (s)**
BAIS2	/	8
U-Net	2.3	5.6
BiAU-Net	3.6	7.9
SAFE	/	11.7
Transformer (Local)	2.7	6.4

Despite its significant advantages in burned area extraction, SAFE also has certain limitations. First, the performance of SAFE depends on the pre-training of the SAM model, which means its extraction effectiveness is largely influenced by the quality and diversity of the base model's training data. If certain specific scenarios or environments are underrepresented in the training data, SAFE's performance in these scenarios may be suboptimal. Secondly, since SAFE is a feature-based segmentation method, it may exhibit oversegmentation or undersegmentation under extreme conditions (e.g., areas where post-fire vegetation and other ground features are highly similar), affecting the

accuracy of extraction results. While SAFE is highly effective during daylight hours, its performance during nighttime is constrained by the reliance on optical sensors, which are less effective without sufficient lighting. This limitation affects the method's ability to detect and monitor wildfires continuously around the clock, highlighting a constraint in its current implementation. Furthermore, SAFE's applicability in regions with diverse topographies and climatic conditions—such as mountainous areas or regions with frequent cloud cover—may also be limited. The model's performance could be reduced in such environments unless additional data sources, like SAR imagery, are incorporated. Additionally, SAFE's computational complexity is relatively high; although its segmentation accuracy surpasses traditional methods, the demand for computational resources can be significant when processing large-scale regional imagery, which may impose higher requirements on hardware capabilities. Moreover, the image quality is critical for SAFE's performance, and issues such as cloud cover or poor atmospheric conditions can affect the quality of both Sentinel-2 and MODIS data. The method's robustness to these challenges may need further improvement, particularly in areas with frequent cloud cover or poor image quality. Furthermore, the proposed burned area algorithm is not immune to limitations and uncertainties, particularly in agricultural areas. The traces left by crop fires in farmland are often small and can easily be confused with harvested or tilled areas in the spectral domain. Future research will focus on improving the performance of burned area detection in such areas, further enhancing recall and accuracy, and ultimately extending its application to fire science. Additionally, exploring transfer learning techniques and multi-modal data integration, such as combining optical and SAR imagery, could improve the method's generalization ability and robustness in challenging environments.

In addition to these limitations, the applicability of SAFE in diverse climatic and topographical conditions remains an important area for further exploration. For example, regions with extreme weather conditions (e.g., frequent heavy rainfall, snow, or cloud cover) may hinder the performance of the optical data, such as Sentinel-2, which could affect both the segmentation accuracy and the method's reliability. In such cases, the integration of other satellite data sources, such as Synthetic Aperture Radar (SAR), which can penetrate clouds and provide consistent coverage, might be essential for improving the robustness of the model. Another potential limitation lies in the dependency on high-quality image data for accurate detection. In regions where imagery suffers from low resolution or poor atmospheric conditions, the performance of the SAFE method might degrade, requiring further optimization to handle noisy or inconsistent data. For future research, optimizing the SAFE method for even better generalization capabilities is a key area of focus. One potential direction could involve the application of transfer learning, where pre-trained models on global wildfire datasets could be fine-tuned for specific regions with limited data, improving the model's performance in data-scarce areas. Additionally, optimizing the algorithm to handle varying topographical features, such as mountainous terrains or dense urban areas, could further extend the model's applicability. Moreover, the incorporation of multi-modal remote sensing data—combining optical, radar, and thermal imagery—could offer significant improvements in detecting burned areas under different environmental and climatic conditions.

6. Conclusions

In this study, we proposed an automated algorithm called SAFE based on the Segmentation Anything Model (SAM) to detect burned areas (BAs) from single-scene imagery. SAFE employs several strategies to effectively overcome the challenges faced by traditional methods. First, we utilized hotspot product data (e.g., MODIS and VIIRS) generated based on a set of optimal spectral indices to address the challenges posed by background het-

erogeneity and the spectral diversity of burned areas. Second, we used single-temporal Sentinel-2 MSI imagery to detect burned area candidates, reducing the complexity and uncertainty associated with multi-temporal data processing. Finally, we integrated the SAM model to effectively refine the burned area candidates, ensuring precision and completeness in the extraction results. Experimental results indicate that SAFE demonstrates an excellent performance in complex environments such as grasslands and forests, significantly outperforming traditional threshold methods like BAIS2 and other deep learning models across multiple test regions. SAFE provides efficient burned area extraction capabilities and can significantly reduce extraction time while maintaining high accuracy, making it particularly suitable for fire monitoring applications that require rapid response. Furthermore, SAFE's unsupervised nature and strong generalization ability ensure a robust performance in different environments, enabling it to handle a wide range of scenarios, including smoke occlusion, illumination variations, and complex backgrounds. Despite its significant advantages in burned area extraction, SAFE also has certain limitations, particularly in specific environments such as agricultural fields, where fire traces can easily be confused with harvested or tilled areas. Future research will focus on optimizing the computational efficiency of SAFE, enhancing its detection performance in specific environments, improving recall and accuracy in burned area extraction, and ultimately extending its application to fire science.

Author Contributions: Conceptualization, S.L.; methodology, S.L. and Y.X.; software, S.L. and H.C.; validation, Y.C. and H.C.; formal analysis, S.L. and H.C.; investigation, S.L. and T.Z.; resources, S.L. and H.C.; data curation, S.L. and H.C.; writing—original draft preparation, S.L.; writing—review and editing, S.L. and Y.X.; visualization, S.L. and H.C.; supervision, S.L.; project administration, S.L.; funding acquisition, S.L. All authors have read and agreed to the published version of the manuscript.

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Data Availability Statement: The data presented in this study are available on request from the corresponding author. The data are not publicly available due to the fact that the research group's wildfire-related research is still being carried on; the later work will also rely on the current dataset to carry out.

Conflicts of Interest: The authors declare no conflicts of interest.

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